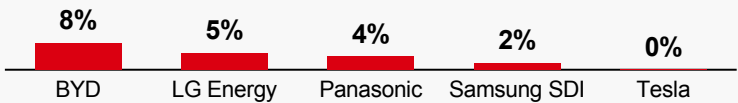


One position's tail is bounded. The other's is not.

CATL is an outright long. Sizing is its only risk control.

Variance of CATL a peer short would remove, %

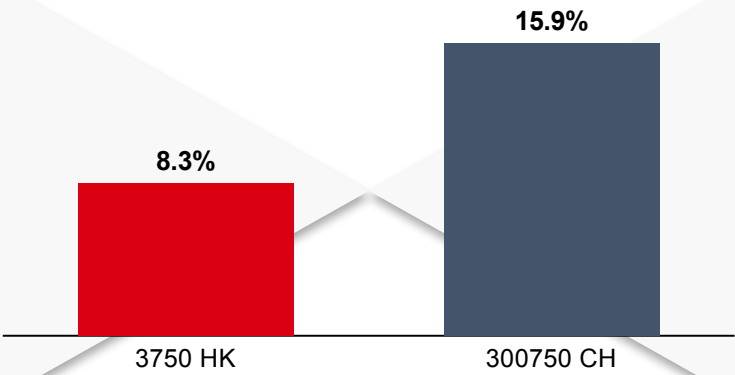
Why there is no peer leg. Minimum-variance hedge ratio, daily since listing, against the full 0 to 100 scale.



The best of them removes 8%. Correctly, there is no peer leg. It would have been a second position.

CATL's worst single session, % lost

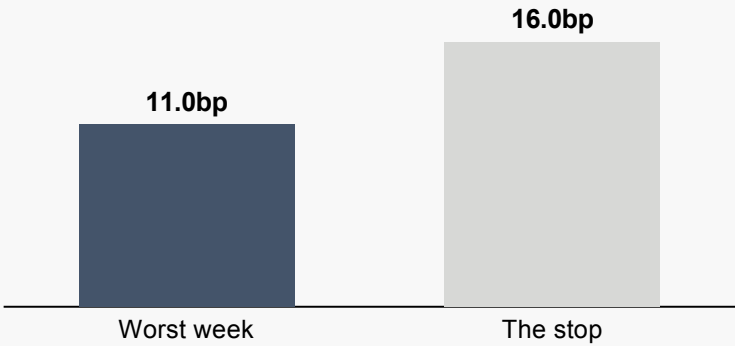
Carried in full. The H share is the traded line and the more volatile of the two, at 46% annualised against 37%.



No stop, no option, no offset. A single session has taken 8.3% off the traded line and 15.9% off the A.

The steepener's stop, tested, bp

Worst weekly flattening observed since 2023 against the 16bp stop. No week in the sample moved through it.



0.69x. This hedge does work. 0 of 186 weeks flattened through 16bp.

Two positions, two different kinds of risk control, and only one of them exists

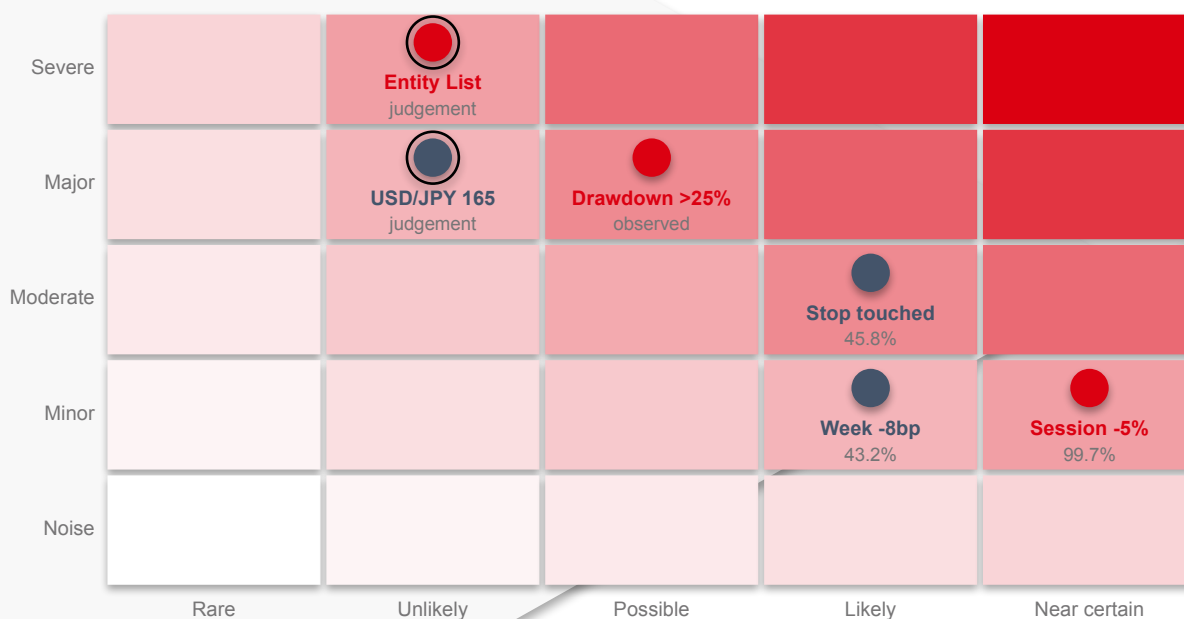
The steepener's loss is bounded by a stop no week in the sample has moved through. CATL's is bounded by nothing, so **position size is the entire risk control**. Size it off the H share's 46% volatility rather than the A share's 37%, because the H share is what is being bought and it is the more volatile line.

Where the risk sits

Six risks, measured. And whether the two trades share them.

Likelihood against impact, six months

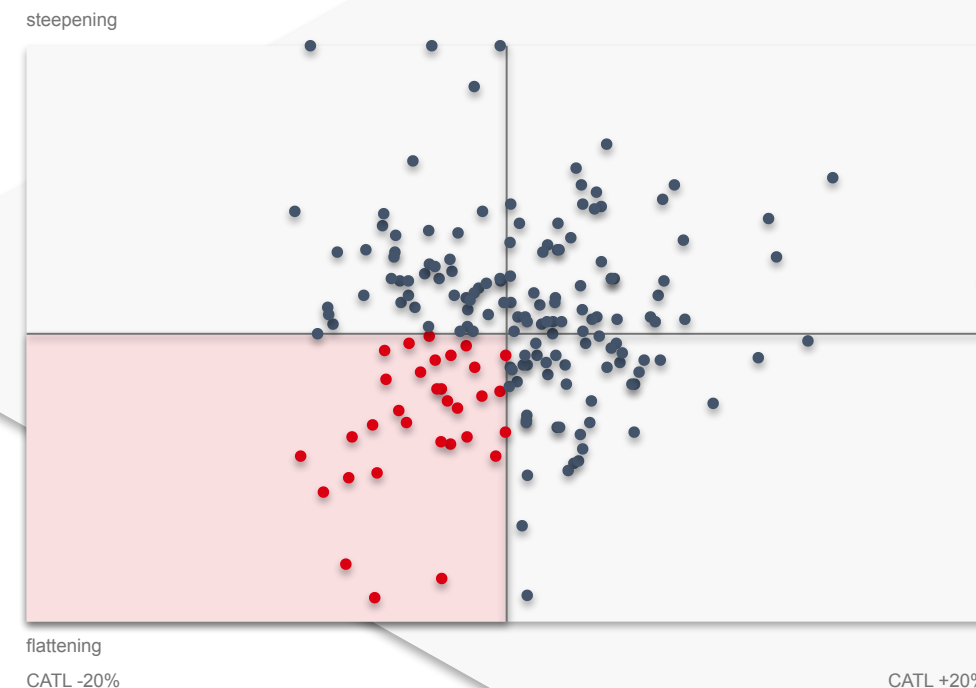
IMPACT



ringed = judgement

LIKELIHOOD, six months

Do they lose together?

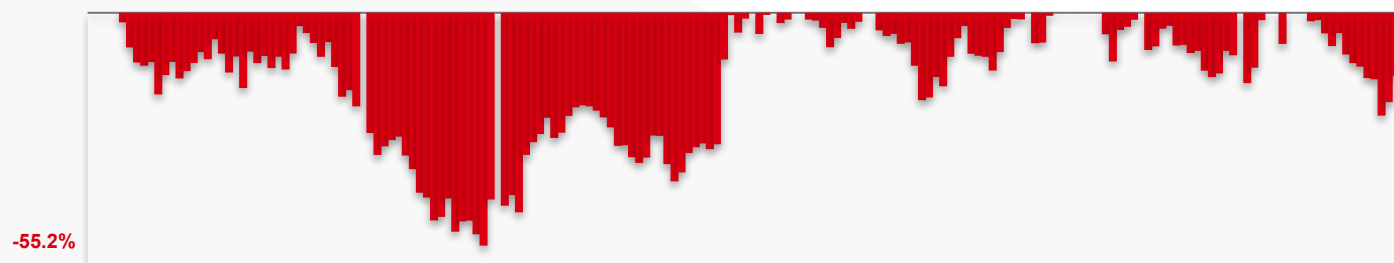


Independent trades. The concentration is inside CATL: 90% of sessions below a prior peak, drawdown 24.7%, no stop.

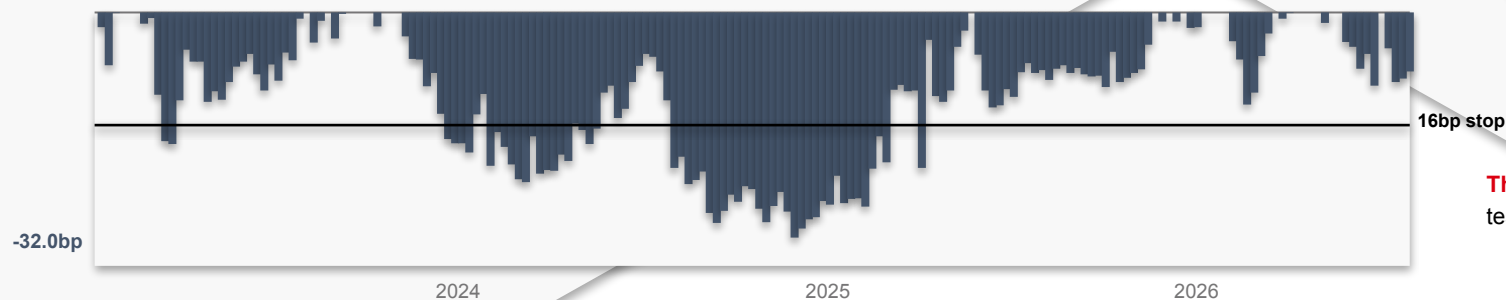
Time underwater

Distance below the running peak, weekly.

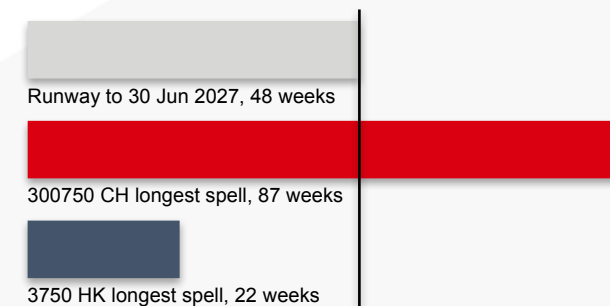
CATL 300750 CH, %



JGB 2s10s steepener, bp



Against the runway



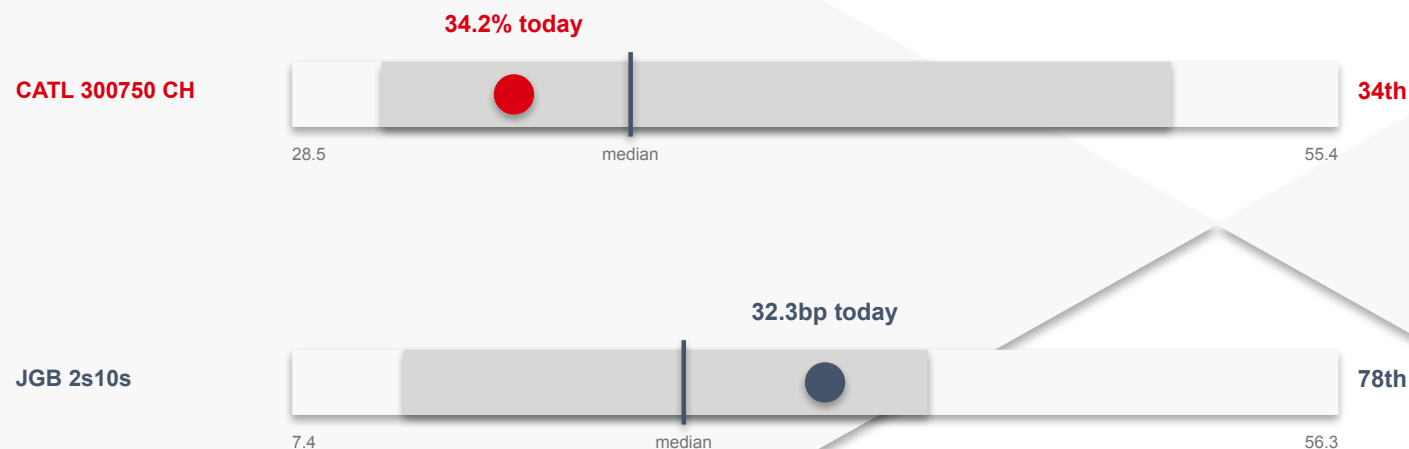
The longest spell is 1.8x the runway. The traded line is younger and has not been tested.

The risk is not a bad week, it is a long one. The 16bp stop is never gapped through, but drift takes it past twice over.

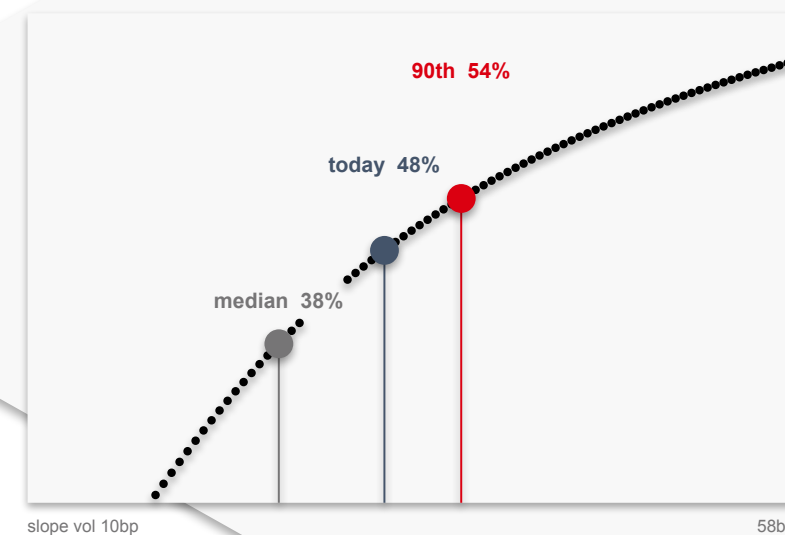
Which sample is calm?

Where each sleeve's volatility sits in its own history.

Annualised volatility, own history



Chance of touching the 16bp stop

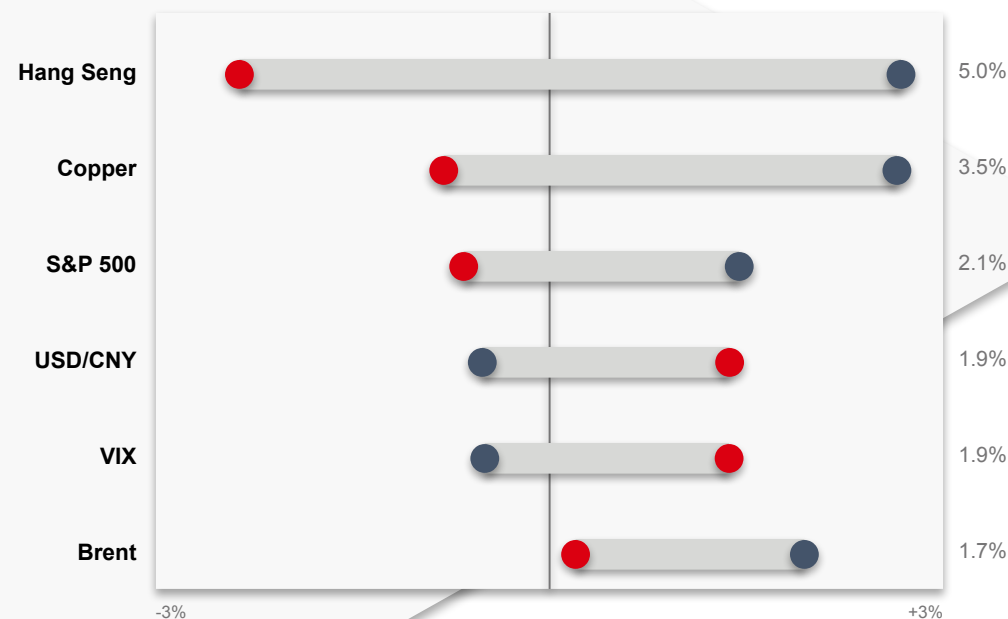


The calm sample is CATL's, not the curve's. Sized at the 34th percentile, it is 1.5x light against a 90th-percentile regime.

What actually moves each sleeve

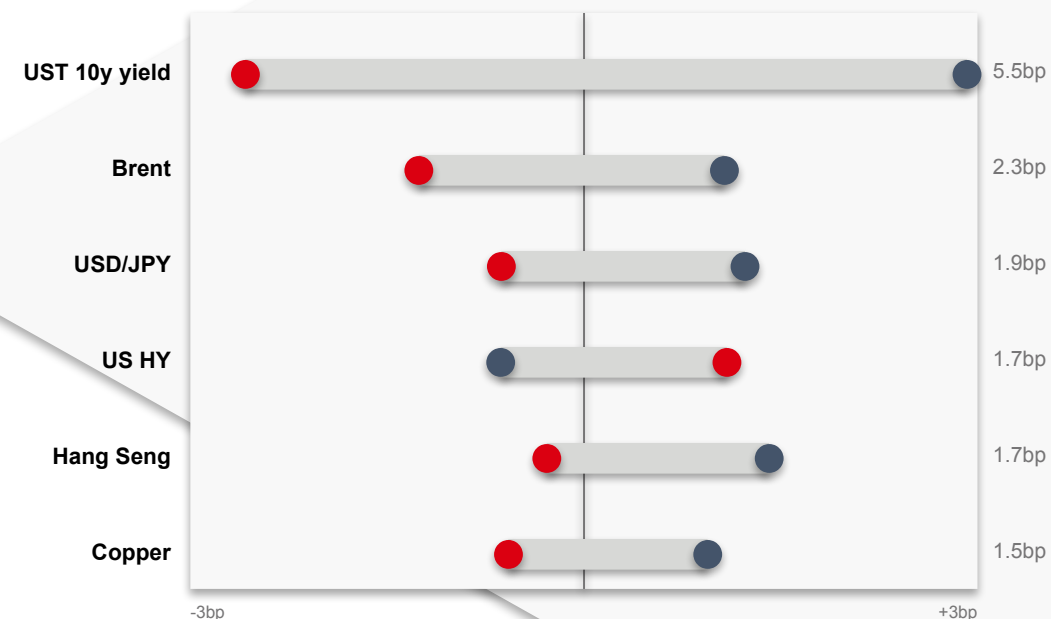
Mean weekly P&L in the bottom and top fifth of each factor's own move.

CATL 300750 CH, weekly %



red = factor's worst fifth navy = its best fifth

JGB 2s10s steepener, weekly bp



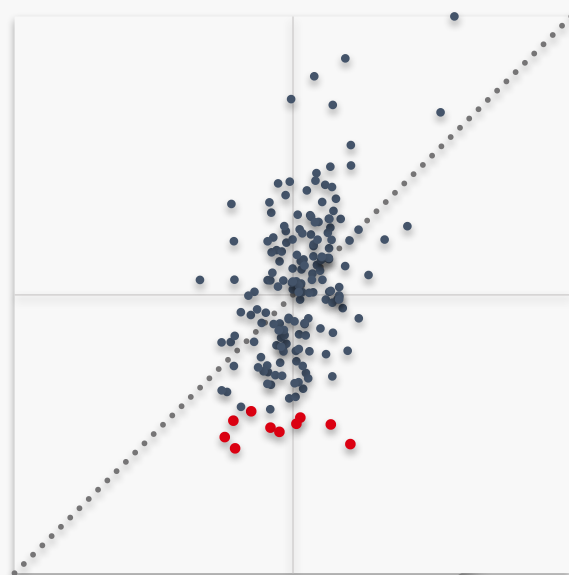
CATL is a Hang Seng position. The steepener's largest driver is the UST 10-year, not the yen.

What a hedge could reach

Eight-factor model. Predicted against actual, weekly.

CATL 300750 CH

actual +16%



22.7%

reachable by
any hedge

12%

of the ten
worst weeks

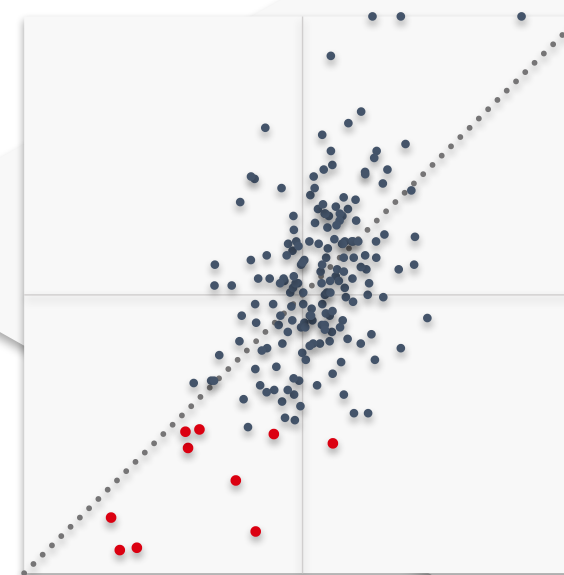
predicted -16%

+16%

red = the ten worst weeks grey line = a perfect model

JGB 2s10s steepener

actual +12bp



31.2%

reachable by
any hedge

53%

of the ten
worst weeks

predicted -12bp

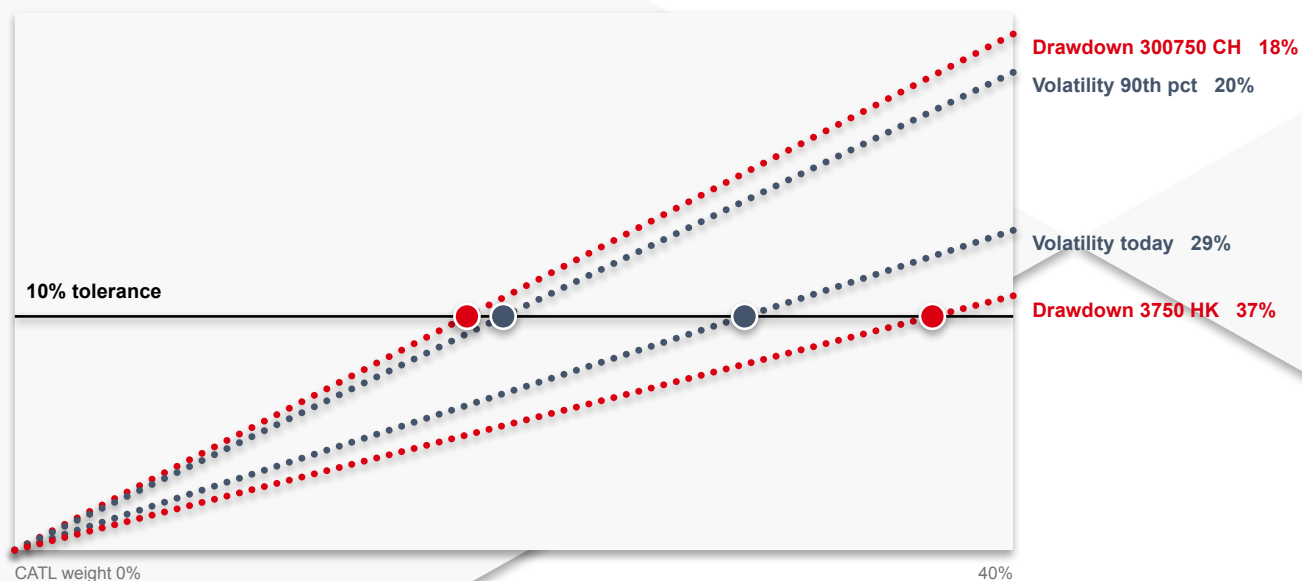
+12bp

Nothing tradeable reaches CATL: factors saw 12% of its ten worst weeks. For the steepener, 53%.

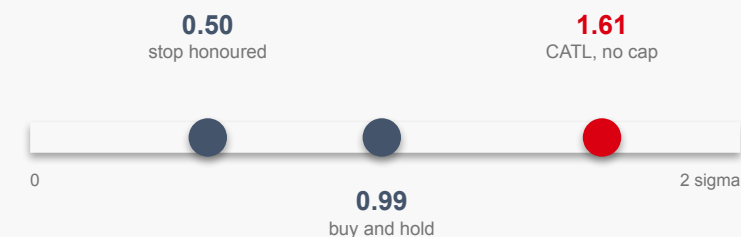
How big can CATL be?

Size is the only control left. This is how much.

book impact, % of risk capital



Worst loss, in own sigma



A stop turns 1.61 sigma into 0.50. CATL has none.

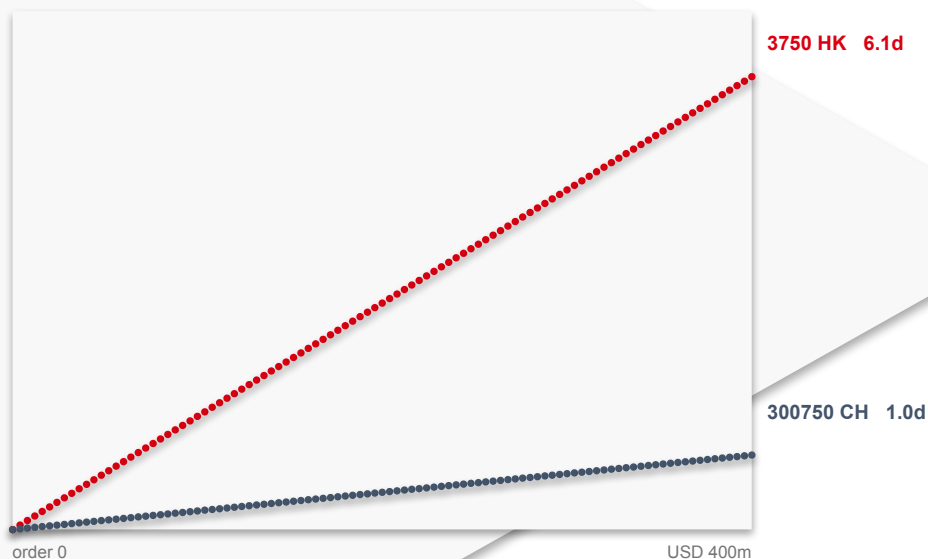
Drawdown binds before volatility. The longer history caps CATL at 18% of risk capital, the traded line's at 37%.

Getting out

Days to exit, and what it costs.

Days to exit at 20% of turnover

trading days



1.7x

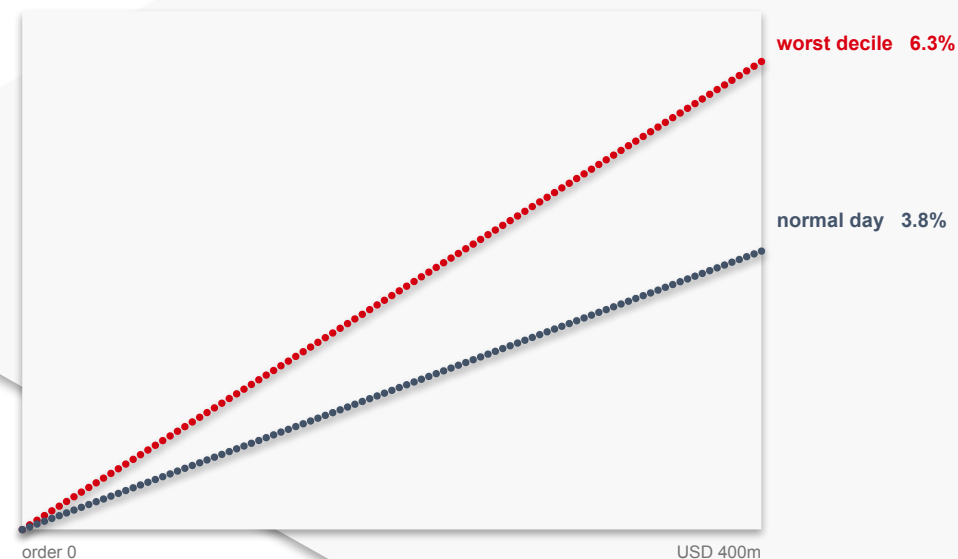
impact on the worst days

6.3x

H share thinner than A

Cost to exit, % given up

price impact, %



14%

of turnover is the H line

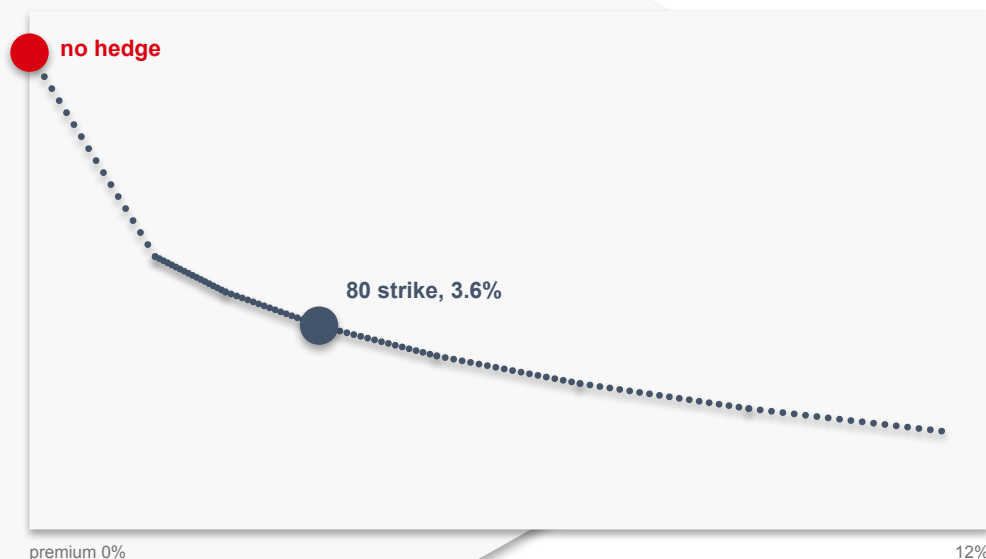
The size is tradeable. The exit is not free, and it gets 1.7x dearer on exactly the days it is wanted.

What bounding the tail costs

Puts to the 30 June 2027 prohibition.

Premium paid against worst case bought

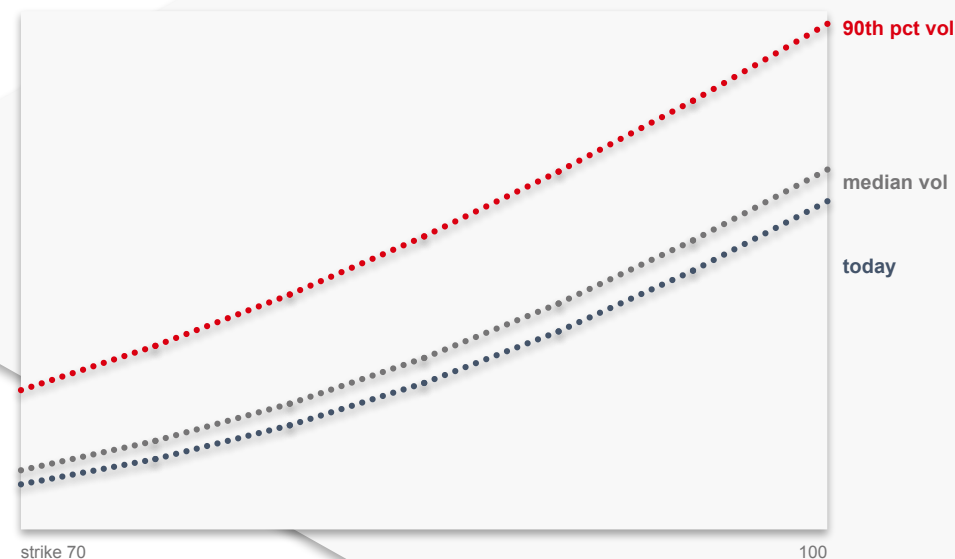
worst case, % of notional



The first 1.6% removes 24 points. Everything after it costs more per point.

And what volatility does to it

premium, % of notional



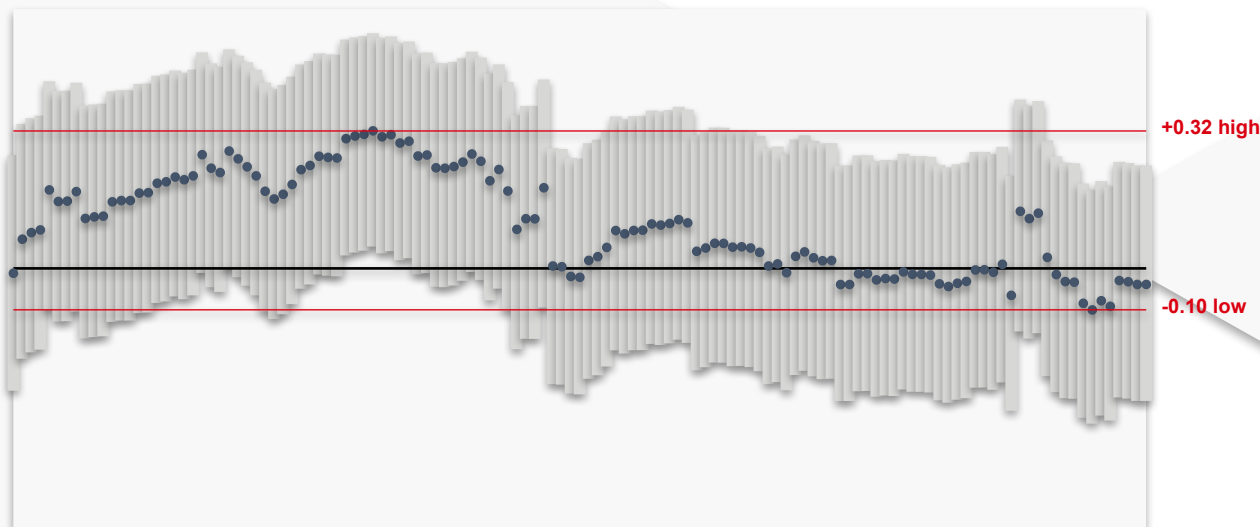
The steepener bounds its tail with a stop, for nothing. CATL's costs 3.6% of notional to cut 55% to 24%.

Is the independence stable?

Rolling 52-week correlation, with its own confidence band.

Correlation between the two sleeves

rho +0.6



rho -0.6

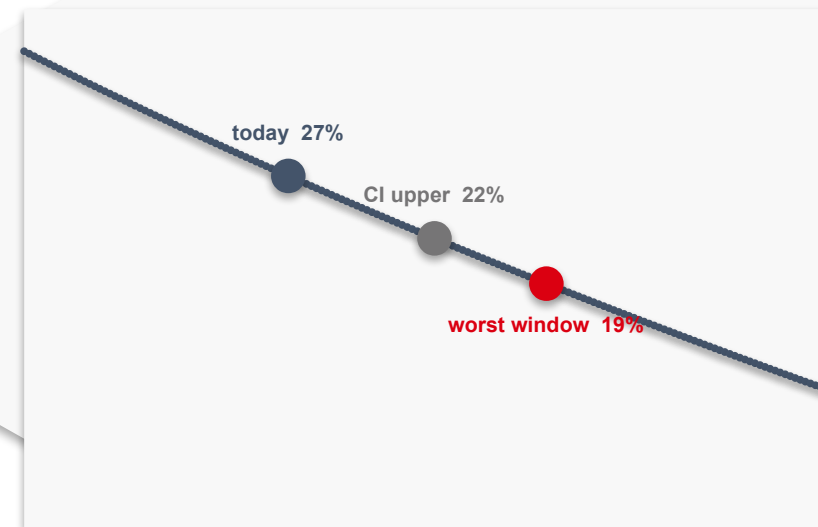
2024-02

94% of windows cannot reject zero. The band is wide because 52 weeks is a small sample.

2026-07

What it is worth

diversification benefit, %



rho -0.2

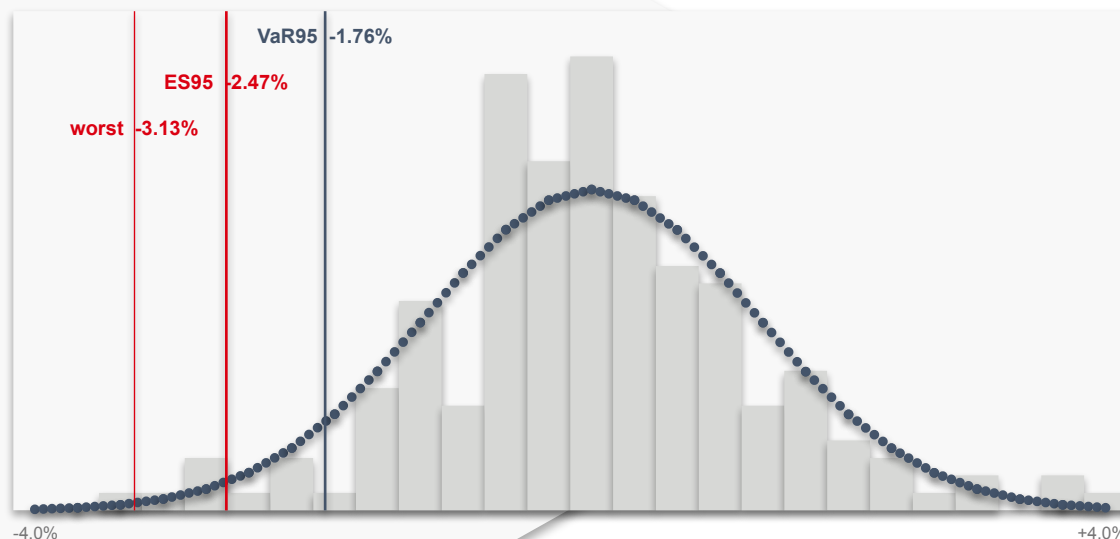
+0.6

Rho has never left -0.10 to +0.32 on a rolling year. The diversification holds, and at its worst it is worth 19% not 27%.

The book, not the sleeves

Weekly P&L at risk parity, against a fitted normal.

Weekly book P&L, % of capital



8.6%

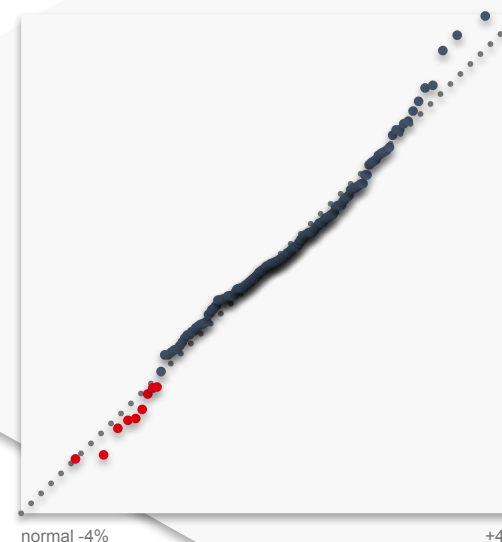
book vol, annualised

1.08x

ES95 against normal

Against a normal

actual +4%



-2.6

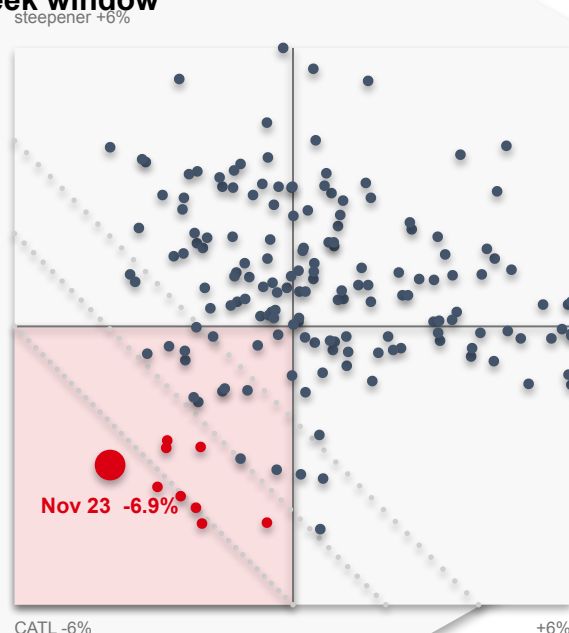
sigma, the worst week

The book's tail is within 8% of normal. That is the sample talking: it has never held a stressed CATL.

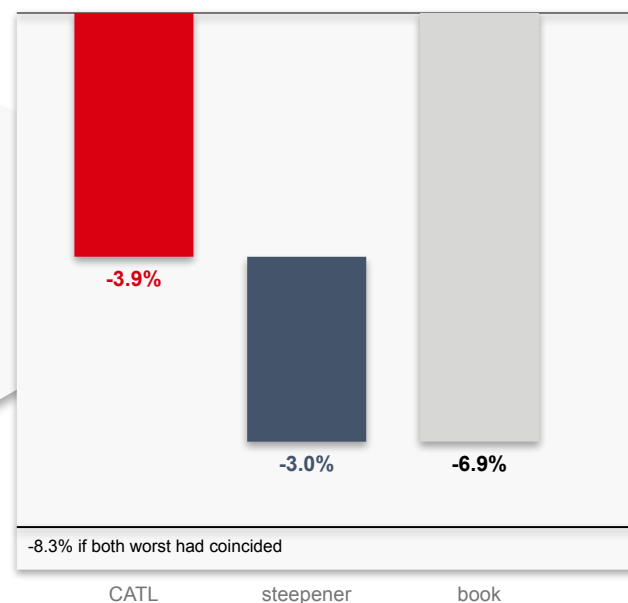
The worst episodes it has actually had

Eight-week windows, and what the other sleeve was doing.

Every 8-week window



The worst one



Who cushions whom

+20%
the steepener offsets
in CATL's worst four

-26%
CATL adds to the loss
in the steepener's worst four

The steepener cushions CATL. CATL does not return the favour, and the worst episode reached 83% of what both losing together could cost.